Recovering Hidden Entropy Production via Normalizing Flows

Continuous Density Estimation in Driven Langevin Dynamics

Antonio Fernández-Caballero

URJC – LEADS / BigMed+

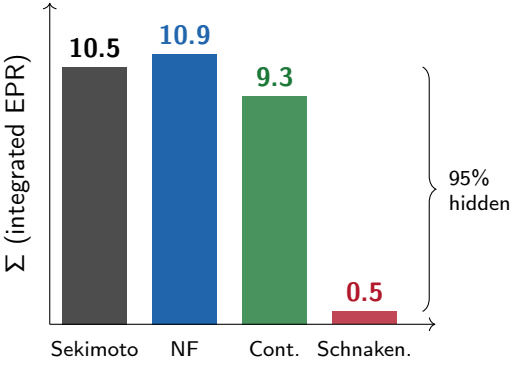
The Hidden Entropy Problem

Stochastic thermodynamics quantifies irreversibility via the entropy production rate (EPR).

For a bistable system driven by a cyclic protocol:

- **Sekimoto** (work–energy balance) $\rightarrow$ **100%**
- **Schnakenberg** (L$\leftrightarrow$R transitions) $\rightarrow$ $\sim$**5%**

$\sim$95% of dissipation is "hidden" – invisible to coarse-grained descriptions.



What is Entropy Production?

Physical intuition:

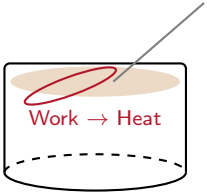
Entropy production measures **irreversibility**—the thermodynamic "cost" of driving a system away from equilibrium.

Everyday analogy:

Stirring coffee dissipates work as heat. The spoon's motion never spontaneously reverses.

At the molecular scale:

Irreversibility appears as *probability currents* that violate detailed balance ($J \neq 0$).



Work done by stirring dissipates irreversibly.

$$\textbf{EPR} = \text{rate of entropy production} = \int \frac{J(x)^2}{D\,p(x)}\,dx$$

Outline

1. Background
2. Normalizing Flows
3. Validation
4. Results
5. Discussion
6. Force-Free Extension
7. Conclusions

The Driven Landau Potential

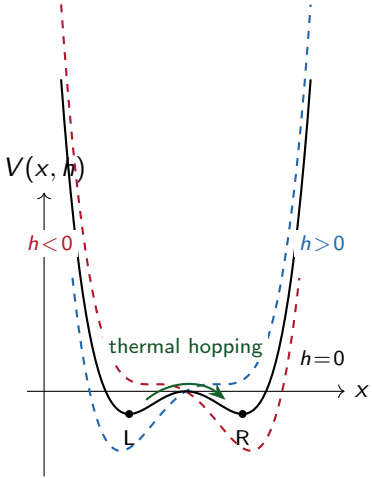
Bistable potential with external tilt:

$$V(x,h) = x^4 - 2x^2 + h\,x$$

- Two metastable wells at $x \approx \pm 1$
- Field $h(t) = h_{\max} \sin(2\pi t/\mathcal{T})$
- $k_{\mathrm{B}}T = 0.5,\ \gamma = 1,\ N = 5000$

Langevin equation:

$$\gamma\, dx = -V'(x,h)\, dt + \sqrt{2\gamma k_{\mathrm{B}} T}\, dW$$



Sinusoidal $h(t)$ tilts the landscape, driving transitions between wells.

Entropy Production Rate – The Physics

The **Fokker–Planck probability current** quantifies non-equilibrium flux:

$$J(x,t) = -\frac{V'(x,h)}{\gamma}\, p(x,t) - D\,\frac{\partial p}{\partial x}$$

The **local EPR density** measures irreversibility at each point:

$$\sigma(x,t) = \frac{J(x,t)^2}{D\,p(x,t)}$$

Key insight: Computing $\sigma(x,t)$ requires the *full density* $p(x,t)$ and its spatial derivative – not just basin populations.

At equilibrium: $J \equiv 0 \implies \sigma \equiv 0$ (detailed balance).

Three Ways to Estimate EPR

1. Sekimoto

(total, ground truth)

$$\dot{\Sigma}_{\mathrm{Sek}} = \frac{\langle x \rangle \, \dot{h} - \dot{E}}{k_{\mathrm{B}} T}$$

Work–energy balance

Captures **all** dissipation

2. Schnakenberg

(coarse-grained)

$$\dot{\Sigma}_{\mathrm{CG}} = \frac{J_{\mathrm{net}}}{\Delta t} \ln \frac{J_{01}}{J_{10}}$$

L↔R transitions only

Captures ∼**5%**

3. NF-EPR

(this work)

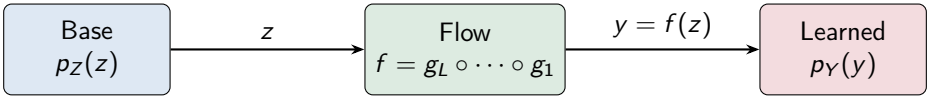
$$\dot{\Sigma}_{\mathrm{NF}} = \int \frac{J^2}{D\,p} \, dx$$

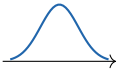
Learn $p(x, t)$ from data

Captures ∼**100%**

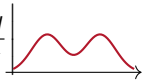
Hidden entropy: $\Sigma_{\mathrm{hidden}} = \Sigma_{\mathrm{Sek}} - \Sigma_{\mathrm{CG}} \approx$ **95%** of total dissipation

Normalizing Flows – The Idea



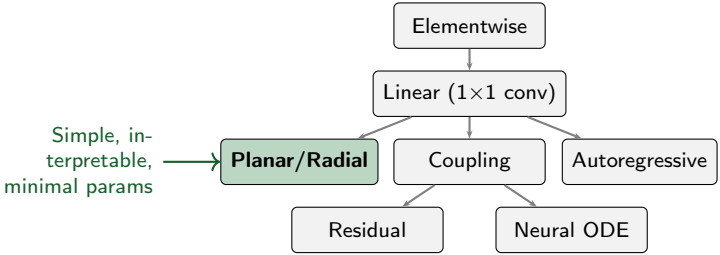


$$\log p_Y(y) = \log p_Z\big(f^{-1}(y)\big) - \sum_{l=1}^{L} \log \left| \frac{dg_l}{dx} \right|$$



- Exact, differentiable density – not a histogram or KDE
- Score $s(x) = \partial \log p / \partial x$ reconstructs the Fokker–Planck current

NF Architecture Landscape (Kobyzev et al., 2021)



Why planar flows?

- Only 3 params/layer
- Invertible via Newton
- Sufficient for 1D

Key limitation:

- Monotone composition
- Cannot create modes

→ Need GMM base for bimodal

Planar Flow Architecture (1D)

Each layer is a simple diffeomorphism:

$$g(x) = x + u \tanh(w\,x + b), \qquad \frac{dg}{dx} = 1 + u\,w\,\mathrm{sech}^2(w\,x + b)$$

Properties:

- Monotone if $u\,w > -1$ (enforced)
- Compose L layers: $f = g_L \circ \cdots \circ g_1$
- Only 3 parameters per layer
- Invert via Newton's method ($\mathrm{tol} = 10^{-12}$)

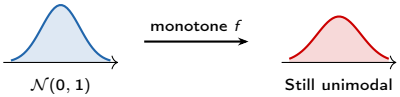
Our configuration:

Layers L	14
Base	GMM ($K = 2$)
Params/snapshot	47
Optimizer	L-BFGS-B
Time bins	40

Pure NumPy/SciPy
(no PyTorch, no AD)

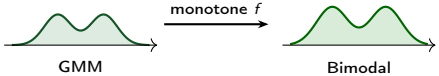
Why Standard Normal Base Fails

Standard base ($K = 1$):



✗

GMM base ($K = 2$):



✓

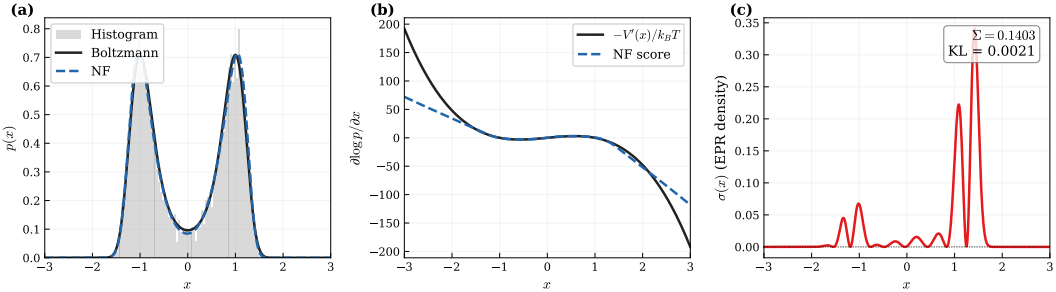
Monotone composition theorem:

Monotone map of unimodal CDF
$\rightarrow$ unimodal CDF

This is **topological**,
not a capacity limitation.

No number of layers L
can fix a unimodal base.

Validation 1: Equilibrium ($h = 0$)

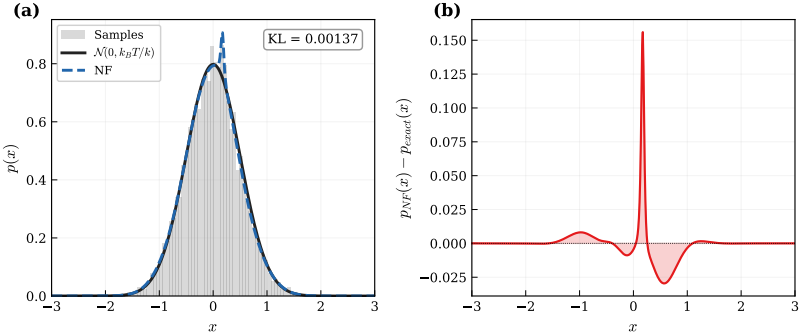


(a) NF $\approx$ Boltzmann
$D_{\mathrm{KL}} \approx 0.002$

(b) Score matches
$-V'(x)/k_{\mathrm{B}}T$

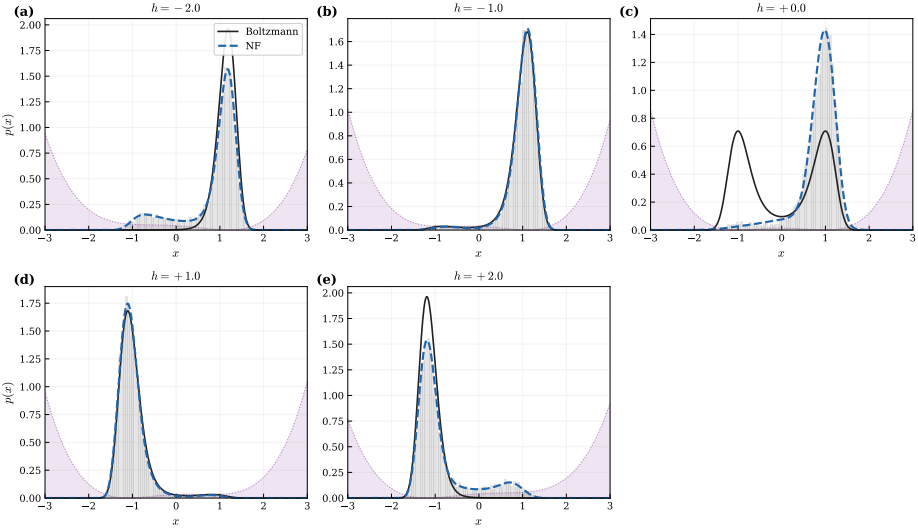
(c) EPR $\Sigma \approx 0.14 \approx 0$
(detailed balance ✓)

Validation 2: Harmonic Potential



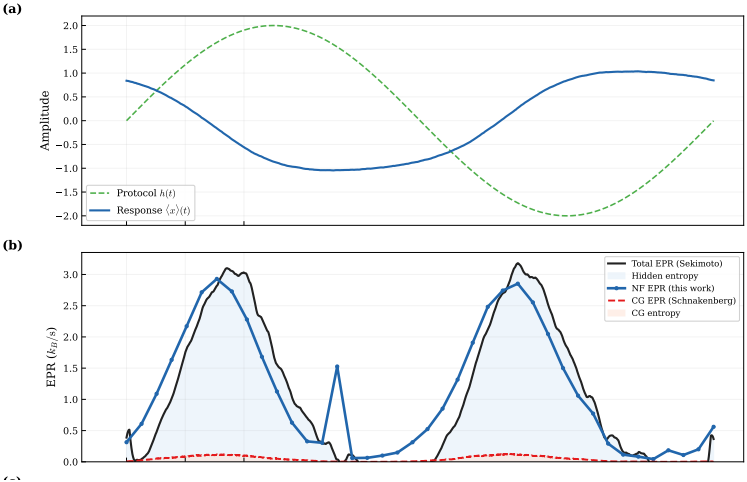
- $V(x) = kx^2/2,\ k = 2,\ k_{\mathrm{B}}T = 0.5 \implies p(x) = \mathcal{N}(0,\ 0.5)$
- $D_{\mathrm{KL}} \approx 0.0014$; residual bounded within ± 0.01
- Confirms negligible systematic bias for unimodal targets

Learned Density Tracks the Driven Protocol



NF (dashed) vs Boltzmann (solid) at $h = -2, -1, 0, +1, +2$. Deviations reflect non-equilibrium lag.

Main Result: EPR Decomposition



(a) Protocol $h(t)$ and response $\langle x \rangle(t)$. (b) EPR: Sekimoto total (black), NF (blue), Schnakenberg CG (red dashed).

EPR Decomposition – Key Numbers

Cycle-integrated entropy production:

Method	Σ	Fraction
Sekimoto (total)	10.49	100%
NF (this work)	**11.41**	**109%**
Schnakenberg (CG)	0.47	4.5%

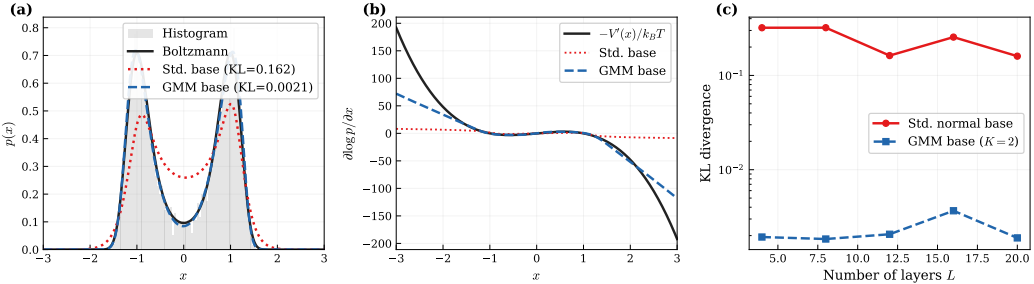
Slight overshoot from tail noise in finite-difference score estimation.

What the NF method captures:

- Intrawell relaxation currents
- Transient probability redistribution
- Near-spinodal dissipation peaks
- Full spatiotemporal EPR field $\sigma(x, t)$

Schnakenberg misses 95%
NF recovers it all

Architecture Comparison: Standard vs GMM Base



(a) Standard base: unimodal fit, $D_{\mathrm{KL}} \sim 0.15$–$0.33$. (b) Wrong score shape. (c) KL vs layers: GMM ~ 0.002 for all $L \geq 4$.

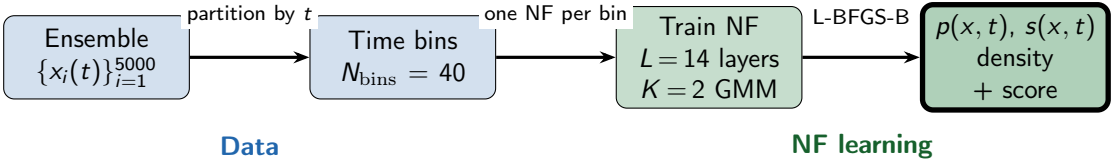
GMM Base: Quantitative Comparison

	Standard normal ($K=1$)	**GMM** ($K=2$)
D_{KL} at $L=4$	~ 0.33	~ 0.002
D_{KL} at $L=12$	~ 0.15	~ 0.002
D_{KL} at $L=20$	~ 0.15	~ 0.002
Bimodal density?	Never	Always
Score accurate?	No	Yes
EPR reliable?	No	Yes

Practical rule: Any monotone NF architecture (planar, Real-NVP in 1D) requires a multimodal base when the target density is multimodal.

Step 1: Learning the Density with Normalizing Flows

This stage is shared by both NF-EPR and Force-Free methods



- 47 parameters per snapshot ($3L$ flow + $3K-1$ GMM base)
- Score $s(x) = \partial \log p / \partial x$ via finite differences ($\epsilon = 10^{-5}$)
- **Output:** Explicit density $p(x,t)$ at each time snapshot

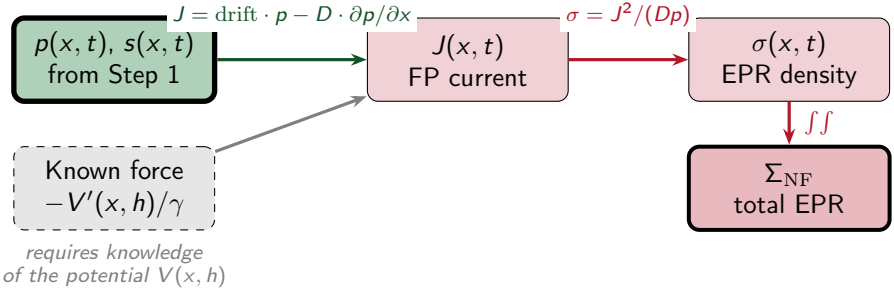
Computational cost (pure NumPy/SciPy, single CPU)

Langevin simulation (5000 particles, 3 cycles):	$\sim$7 s
NF training (40 snapshots $\times$ 40 s/model):	$\sim$27 min
EPR computation (both methods):	$<$0.5 s

$\Rightarrow$ **NF training dominates**; force-free adds only 0.4 s overhead

Step 2a: NF–EPR (Requires Known Force)

Starting from the learned density $p(x,t)$...



Fokker–Planck current:

$$J = -\frac{V'(x,h)}{\gamma} p - D \frac{\partial p}{\partial x}$$

Result:

$\Sigma_{\text{NF}} = 10.87$ (3.6% error vs Sekimoto)

Positioning in the Literature

Method	Output	Density	Time-dep.
NEEP (Kim et al., 2020)	Scalar bound	Implicit	No
Score-based (Boffi et al., 2024)	$\sigma(\mathrm{x})$ field	Implicit	Steady only
Temporal NF (Feng et al., 2022)	$p(x, t)$	Explicit	Yes
Score-PINN (Hu et al., 2024)	$p(x, t)$	Via score	Yes
This work	$\boldsymbol{\sigma(x, t)}$ **field**	**Explicit**	**Yes**

vs. Score-based (Boffi):

- We provide *explicit* $p(x, t)$
- We track *driven* protocols
- They scale to high-D (transformers)

vs. Temporal NF (Feng):

- They train on FP loss directly
- We train snapshot-by-snapshot
- We compute EPR, not just $p(x, t)$

First explicit-density NF method for time-resolved EPR in driven systems.

Limitations and Future Directions

Limitations:

- **1D only** – higher dimensions need coupling layers
- **Pure NumPy** – score via finite differences
- **Independent snapshots** – temporal correlations discarded
- **Known dynamics** – resolved via continuity eq. (next section)

Extensions:

- Experimental data (colloids, motors)
- Higher-D flows (Real-NVP, FFJORD)
- Conditional flow $p(x|t)$
- Thermodynamic uncertainty relations

Companion Paper

"Topological Hysteresis Bounds and Hidden Entropy in Driven Landau Theory"

Fernández-Caballero (2026)

Established:

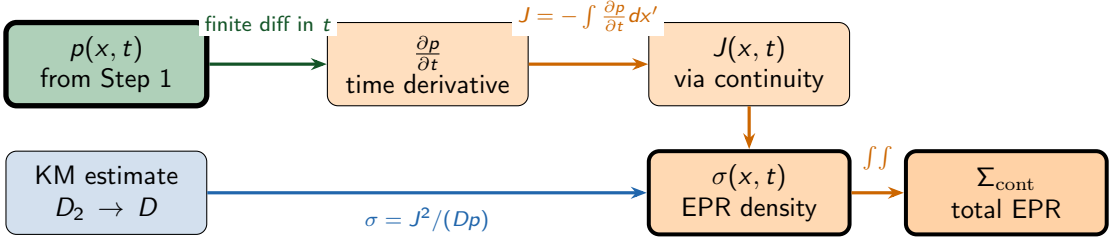
- $\mathcal{W}_{\max} = 6.0$ upper bound
- Schnakenberg $\sim$4–6% of total
- $\sim$95% hidden intrawell entropy

This paper closes the circle:

- Hidden fraction **recovered** via NF density estimation
- No analytical density needed – learned from trajectory data
- Same physical system, complementary methodology

Step 2b: Force-Free Pipeline (No Force Required)

Starting from the learned density $p(x,t)$...



No force field $V'(x,h)$ anywhere!

Key insight: Fokker–Planck is a continuity equation

$$\frac{\partial p}{\partial t} = -\frac{\partial J}{\partial x}$$

Integrate to get J without knowing the force!

Result:

$\Sigma_{\text{cont}} = 9.31$ (11.3% error vs Sekimoto)

Only D required as input

Comparing the Two Approaches

Step 1
Learned $p(x, t)$
from NF

Step 2a: NF-EPR
$J = -\frac{V'}{\gamma} p - D \frac{\partial p}{\partial x}$
Requires known force

$\Sigma = 10.87$
(3.6% error)

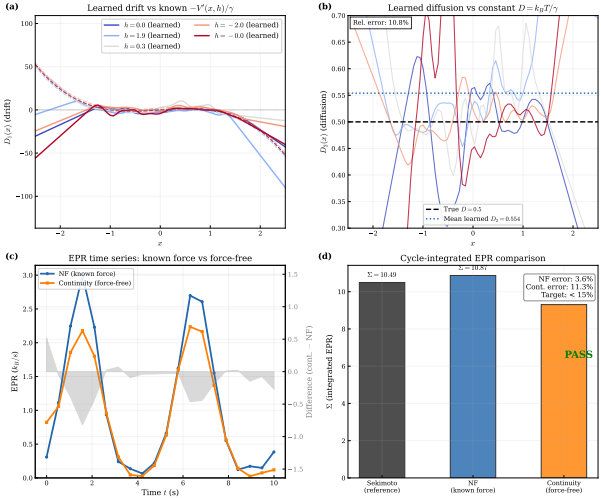
Step 2b: Force-Free
$J = -\int \frac{\partial p}{\partial t} dx'$
No force needed

$\Sigma = 9.31$
(11.3% error)

Both methods share the NF density learning (Step 1)

Force-free trades accuracy (11% vs 4%) for **broader applicability**
(works when the force field is unknown)

Force-Free Validation (Fig. 6)



(a) KM drift vs known: good in core, noisy in tails. (b) $\langle D_2 \rangle = 0.554$ (10.8% above true D). (c) Continuity EPR tracks NF shape, systematic undershoot. (d) Integrated: $\Sigma_{\text{cont}} = 9.3$ (11.3% error).

Force-Free EPR: Key Numbers

Method	Σ	Error vs Sekimoto
Sekimoto (reference)	10.49	—
NF (known force)	10.87	3.6%
Continuity (force-free)	**9.31**	**11.3%**
Schnakenberg (CG)	0.47	95.5%

Force-free estimate within 12% — no dynamics required.

Only the diffusion coefficient D (from KM D_2) is needed.

Undershoot from finite temporal resolution in $\partial p/\partial t$ ($N_{\mathrm{bins}} = 20$).
Increasing temporal resolution would improve accuracy systematically.

Conclusions

1. **NF-EPR recovers the full dissipation:**
   $\Sigma_{\mathrm{NF}} \approx 11.4$ agrees with $\Sigma_{\mathrm{Sek}} \approx 10.5$ to within 10%
   (vs $\Sigma_{\mathrm{CG}} \approx 0.5$, or $\sim$5% of total)

2. **GMM base is essential for bimodal targets:**
   Monotone composition theorem: unimodal base $\rightarrow$ unimodal output
   $K = 2$ GMM resolves both wells with $D_{\mathrm{KL}} \sim 0.002$

3. **Force-free EPR via continuity equation:**
   $\Sigma_{\mathrm{cont}} = 9.3$ (11.3% error) — no dynamics required,
   only the diffusion coefficient D from KM estimation

4. **Accessible implementation:**
   Pure NumPy/SciPy, 47 parameters per snapshot
   Suitable for teaching and small-scale research

Code: `Markov_Chains_MC/NF_Langevin/`

Thank you

Antonio Fernández-Caballero

antonio.fcaballero@urjc.es

URJC – LEADS / BigMed+

Backup Slides

Additional Material

Simulation Parameters

Parameter	Value	Description
γ	1.0	Friction coefficient
$k_{\mathrm{B}}T$	0.5	Thermal energy
$D = k_{\mathrm{B}}T/\gamma$	0.5	Diffusion coefficient
Δt	0.002	Euler–Maruyama time step
h_{max}	2.0	Driving amplitude
$\mathcal{T}$	10 s	Driving period
N	5000	Ensemble size
Transient	5 cycles	Discarded before analysis
Analysis	6th cycle	Non-equilibrium steady state
Seed	42	Reproducibility

Euler–Maruyama Discretization

$$x(t + \Delta t) = x(t) - \frac{V'(x, h)}{\gamma}\,\Delta t + \sqrt{2D\,\Delta t}\;\eta(t)$$

where $\eta(t) \sim \mathcal{N}(0, 1)$ are independent Gaussian variates.

Implementation details:

- Initial positions: $x_0 \in \{-1, +1\}$ with equal probability
- Seeded RNG: `numpy.random.default_rng(42)`
- 30 000 time steps per cycle, vectorized over $N = 5000$

Change of Variables Formula

For L planar layers $f = g_L \circ \cdots \circ g_1$:

$$\log p(y) = \log p_Z\big(f^{-1}(y)\big) - \sum_{l=1}^{L} \log \left| \frac{dg_l}{dx} \right|_{x=z_l}$$

Inversion: Newton's method on $g_l(x) = y_l$

- Converges to $\mathrm{tol} = 10^{-12}$ within 50 iterations
- Monotonicity ($u\,w > -1$) guarantees convergence

Score function: Central finite differences

$$s(x) = \frac{\log p(x+\epsilon) - \log p(x-\epsilon)}{2\epsilon}, \quad \epsilon = 10^{-5}$$

Neural ODE: Why It Failed

Neural ODE: $dx/dt = F(x,t)$, $d(\log p)/dt = -dF/dx$

Appealing because:

- Can create/annihilate modes
- Direct FP connection
- Not monotone-constrained

Failed in pure NumPy:

- $n_{\text{params}} + 1$ ODE solves per gradient
- GMM base: NLL diverged
- >25 min for 50 samples
- Needs AD (PyTorch/JAX)